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Module 5 Introduction

Lesson 1: Kernel Smoothers

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Module 5 Summative Assessment

Graded Assignment: Module 5 Summative Assessment

Module 5 Summary

Your grade: 100%

Your best: 100% · Your highest: 100%

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Module 5 Summative Assessment

Instructions

1. Why is the choice of distance metric important in KNN?

Ready to review what you've learned before starting the assignment? I'm here to help.

☒ Different metrics can influence how the nearest neighbors are determined.

☐ It determines the number of neighbors used in the model.

☐ It dictates the type of problem KNN can solve.

☐ It is the primary factor influencing the model's accuracy.

Correct

Due

Attempts

Correct: The choice of distance metric affects the selection of nearest neighbors in KNN.

Submitted Nov 19, 2024 10:02

Try again

2. In which scenario would local regression be particularly useful?

To earn you need at least 80%. We keep your highest score.

☐ When all variables in the dataset are categorical.

☐ When the data is uniformly distributed with a clear global trend.

☐ When the primary goal is to reduce the dimensionality of the dataset.

☒ When the data exhibits varying patterns or trends in different regions.

Correct

View submission

See feedback

Correct: Local regression is useful when the data shows local patterns across different regions.

3. What is a common challenge when using KNN for large datasets?

To earn you need at least 80%. We keep your highest score.

☐ It requires a fixed number of features for all data points.

☒ The algorithm can become computationally intensive as it requires distance calculations for each query.

☐ It tends to overfit regardless of the dataset size.

☐ It automatically reduces the dimensionality of the data.

Correct

Correct: KNN's need to compute distances for each query can be demanding for large datasets.

4. In kernel smoothing, what is the effect of using a larger bandwidth?

To earn you need at least 80%. We keep your highest score.

☒ It results in a smoother estimate but may oversmooth important features in the data.

☐ It reduces the computational complexity of the smoothing process.

☐ It creates a more complex model with more peaks and valleys.

☐ It increases the sensitivity of the smoother to outliers.

Correct

Correct: Larger bandwidths in kernel smoothing produce smoother curves but can oversmooth.

5. In the context of KNN, what is meant by 'feature scaling'?

To earn you need at least 80%. We keep your highest score.

☐ Selecting the most important features for the model.

☐ Reducing the number of features used in the model.

☐ Transforming categorical features into numerical ones.

☒ Adjusting the range of different features to make them comparable.

Correct

Correct: Feature scaling in KNN ensures that all features contribute equally to the distance calculation.

6. What is a potential drawback of local regression?

To earn you need at least 80%. We keep your highest score.

☐ It requires large amounts of data to perform any fitting.

☒ It can be sensitive to the choice of bandwidth, affecting the smoothness of the fit.

☐ It is computationally less intensive than global regression models.

☐ It always results in linear relationships between variables.

Correct

Correct: The choice of bandwidth is crucial and can greatly affect the outcome of local regression.